IB PHYSICS E.5 FUSION AND STARS

Nuclear fusion

- **Fusion = joining of light nuclei → heavier nucleus + energy release**
- Overcomes **Coulomb repulsion** via extremely high temperature, density, and pressure.
- **Energy released** due to increase in **binding energy per nucleon** after fusion.
- Typical stellar fusion:

$$^{2}_{1}H + ^{3}_{1}H \rightarrow ^{4}_{2}\mathrm{He} + ^{1}_{0}n + 17.6\mathrm{MeV}$$

Example:

In the Sun, proton-proton chain dominates → 4 H → 1 He + energy.

Stellar evolution and role of mass

- **Main sequence:** star fuses hydrogen in core → stable.
- **Low/medium mass (< 8 M⊙):** red giant → planetary nebula → **white dwarf**.
- **High mass (> 8 M⊙):** red supergiant → fusion up to Fe → core collapse → **supernova** → neutron star / black hole.
- **Mass determines lifespan** and **end state** — massive stars burn fuel faster.

Hertzsprung-Russel (HR) diagram

- **X-axis:** Temperature (K) increases to the **left**.
- **Y-axis:** Luminosity (relative to Sun) increases upward.
- **Main features:**
  - **Main sequence:** diagonal band (H fusion in cores).
  - **Red giants/supergiants:** upper-right (cool, luminous, large).
  - **White dwarfs:** lower-left (hot, dim, small).
- **Star position** on diagram reveals temperature, luminosity, size, and stage of evolution.

Note!

Luminosity depends on **temperature + surface area** via Stefan-Boltzmann law.

Stellar parallax

- **Parallax = apparent shift** in star's position when observed from Earth **6 months apart**.
- **Parallax angle (p)** = half total angular shift.
- **Distance (d) in parsecs:**

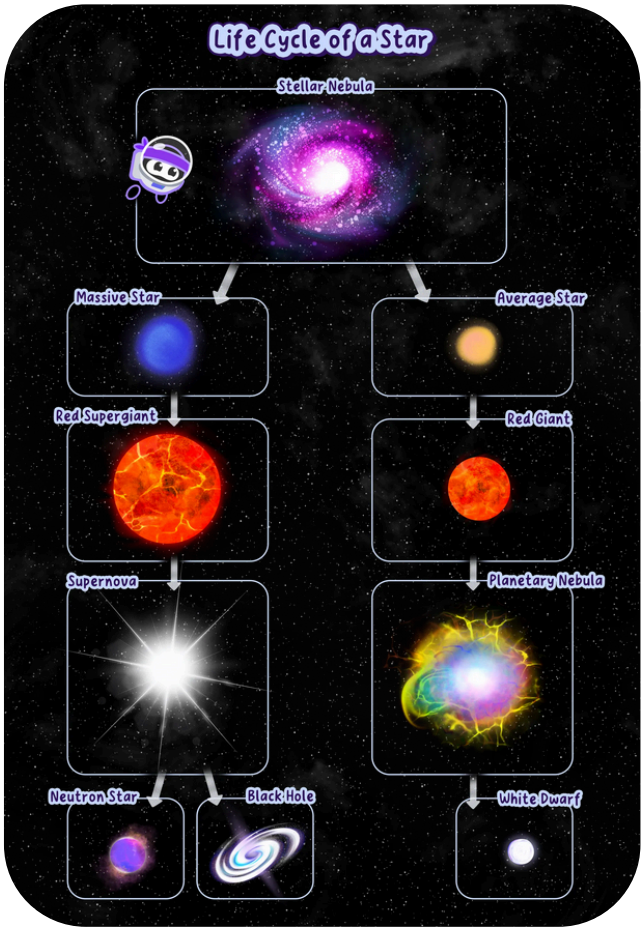
$$d = \frac{1}{p}$$

where p in arcseconds.

- **1 parsec ≈ 3.26 light years ≈ 3.09 × 10^{16} m.**

Conditions for fusion in stars

- **High temperature (10^7–10^8 K):** gives nuclei enough kinetic energy to approach each other.
- **High density:** ensures frequent collisions between nuclei.
- **High pressure (gravitational):** compresses plasma, increases collision probability.
- **Confinement time:** long enough for reactions to occur before nuclei disperse.
- **Quantum tunneling:** allows fusion to occur at lower energies than classical Coulomb barrier.



Stellar luminosity and radius

- **Stefan-Boltzmann Law:**

$$L = 4\pi R^2 \sigma T^4$$

- Relates a star's **luminosity (L)**, **radius (R)**, and **surface temperature (T)**.
- Can rearrange to solve for R or T.

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